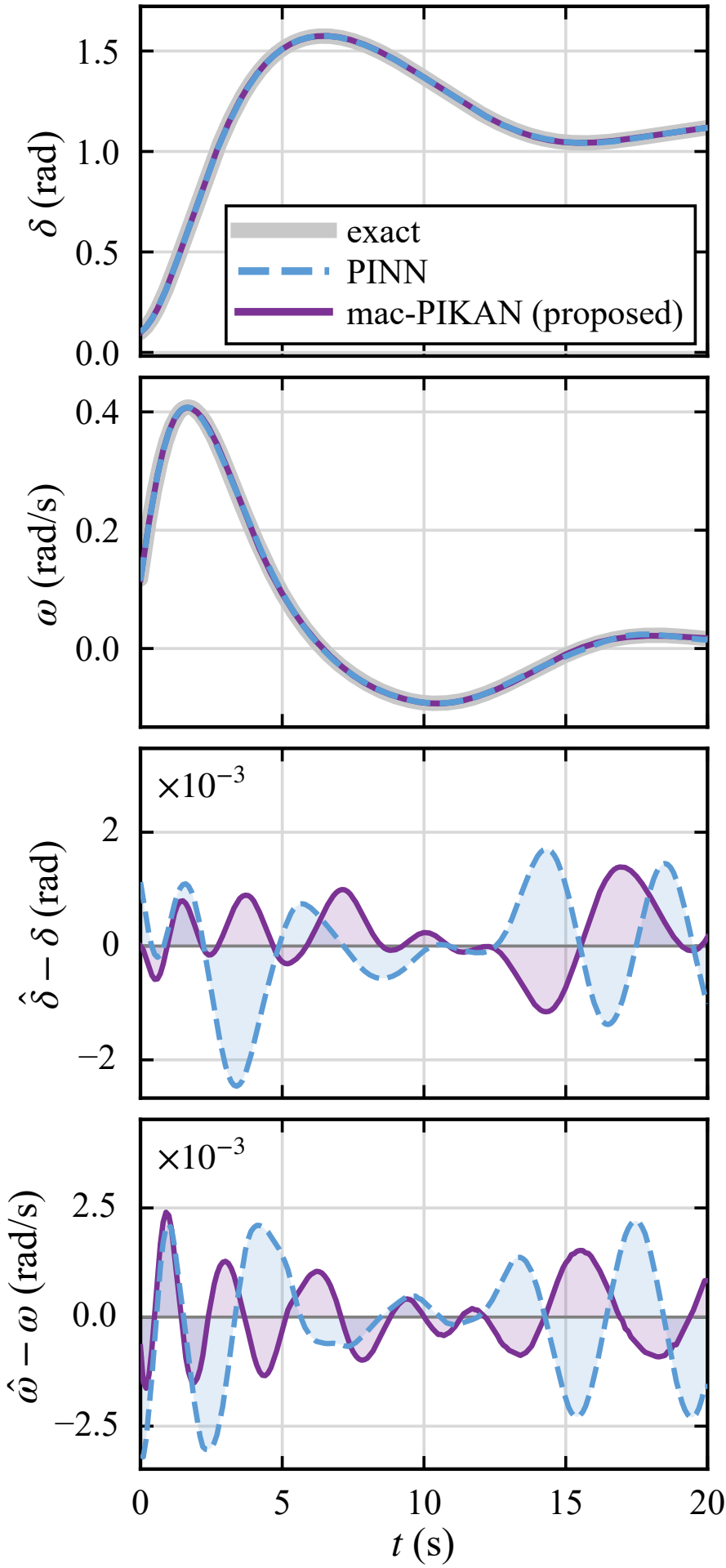

1.5
1.0
0.5
0.0
δ (rad)
exact
PINN
mac-PIKAN (proposed)
0.4
0.2
0.0
ω (rad/s)
×10−3
2
0
−2
δ̂ − δ (rad)
×10−3
2.5
0.0
−2.5
ω̂ − ω (rad/s)
0
5
10
15
20
t (s)